Process Characterization Plan

A-Mab Drug Substance — Cation Exchange Chromatography (Step 7) — PCP-007

Process Development / Manufacturing Science & Technology (MSAT)

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Reviewer	Analytical Development	—	—	—
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Abbreviations

ALCOA+, attributable, legible, contemporaneous, original, accurate and complete; AMV, analytical method validation; CEX, cation exchange chromatography; CV, coefficient of variation; DNA, deoxyribonucleic acid; DoE, design of experiments; ELISA, enzyme-linked immunosorbent assay; ELN, electronic laboratory notebook; HCP, host cell protein; HMW, high molecular weight; ICH, International Council for Harmonisation; IgG1, immunoglobulin G subclass 1; LIMS, laboratory information management system; MSAT, Manufacturing Science and Technology; NOR, normal operating range; OD, optical density; PAR, proven acceptable range; PD, Process Development; PRESS, predicted residual error sum of squares; QA, Quality Assurance; qPCR, quantitative polymerase chain reaction; RSM, response-surface methodology; SEC-HPLC, size-exclusion high-performance liquid chromatography; SOP, standard operating procedure.

1 Purpose and scope

Cation exchange chromatography is the first of the two polishing steps in the A-Mab purification train. The step uses a strong cation exchange resin in bind and elute mode, and a step elution separates the antibody from the impurities that survive capture and low-pH inactivation (PCR-005, PCR-006). It is the principal aggregate reduction step in the process, and it also provides substantial clearance of host cell protein, residual DNA and leached Protein A.

No step downstream of cation exchange reduces aggregate. Anion exchange (PCR-008) and small-virus retentive filtration (PCR-009) act on impurities and on virus, and the ultrafiltration and diafiltration step (PCR-010) concentrates the product and exchanges the buffer, so none of the three removes high molecular weight species. The aggregate content of the drug substance is set by what this step delivers, and the study defined here has to support that claim across the conditions the commercial process may see.

This plan defines the process characterization study for the step. It states the parameters to be studied, the ranges over which they will be varied, the analytical methods, the statistical analysis and the criteria by which the results will be judged. The study will be executed on a qualified scale-down model at the sending site, and its conclusions will be applied to the commercial process at the receiving site through that model. Results, parameter classifications and the operating region will be reported in PCR-007.

The parameters in scope are the 5 listed in Table 6.1, which are the parameters the risk assessment assigned to this step for characterization (RA-001). Resin type, column dimensions, buffer compositions and the collection start threshold are fixed by the platform and by SOP-2010, and are held constant in every run. Resin life-cycle, packing quality and sanitization are governed by SOP-2008 and are not characterized here.

Although some clearance of enveloped virus can be demonstrated across a cation exchange step, no viral clearance claim will be made for this step, and no virus spiking runs are included in this plan. The cumulative clearance claim rests on the low-pH hold (PCR-006), the anion exchange step (PCR-008) and the small-virus retentive filter (PCR-009), and it is consolidated in PCMR-001.

2 Objectives

The study has five objectives, listed below in the order in which the sections of this plan address them.

1. To quantify the effect of protein load, load and wash conductivity, elution buffer pH and elution stop collect on pool aggregate, pool host cell protein and step yield, over the ranges given in Table 6.1.
2. To resolve the interaction between protein load and the conductivity of the load and wash buffers that platform experience predicts for host cell protein clearance, together with any further two-factor interaction the screening design finds active.
3. To define the multivariate region over which the attributes this step governs remain within their acceptance criteria, using a response-surface model fitted to the 4 multivariate parameters.
4. To establish a proven acceptable range for each parameter and each governed attribute, both at set-point and with the remaining parameters varying within their normal operating ranges.
5. To provide the evidence on which each parameter will be classified (SOP-4001) and on which this step's contribution to the control strategy will be based.

The study is a Stage 1 process design activity in the sense of the lifecycle approach to process validation (U.S. Food and Drug Administration 2011), and it follows the enhanced development approach of ICH Q8 and ICH Q11 (International Council for Harmonisation 2009, 2012). The methodology, the attribute framework and the classification logic follow the A-Mab case study (CMC Biotech Working Group 2009).

3 Related documents

This plan sits under three parent documents and is paired with one report, and all of them are listed in Table 3.1 with the relationship each bears to this study.

Table 3.1: Related documents in the characterization package.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-007	Process Characterization Report (Cation Exchange Chromatography (Step 7))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

The procedures that govern execution of the study are listed in Table 3.2, and the validated analytical methods are listed separately in the analytical methods section below.

Table 3.2: Procedures governing execution of the study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2010	Operation of the Cation-Exchange Polishing Chromatography Step	SOP
SOP-2008	Chromatography Resin Life-Cycle, Packing and Sanitization	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The A-Mab downstream process is a platform process, and the cation exchange step is operated in the same mode as it is for three earlier humanized IgG1 products (X-Mab, Y-Mab and Z-Mab). The resin chemistry, the buffer system and the step elution are common to all four products. That experience establishes the mechanism of the separation and the conditions under which the step behaves predictably, so this study is expected to confirm and bound platform behaviour and not to discover it.

Aggregate separation on a cation exchanger follows from avidity. A high molecular weight species presents more of the antibody surface to the resin than the monomer does, binds more strongly and elutes later, so the descending edge of the elution peak is enriched in aggregate. Three of the parameters in scope act on that separation, and each acts on a different part of it. Protein load determines how much of the column capacity is occupied and how well the species are resolved, elution buffer pH determines how strongly both species are held at the point of elution, and the stop collect set-point determines how much of the descending edge enters the product pool.

Host cell protein clearance depends on protein load and on the conductivity of the load and wash buffers. Host cell proteins are a heterogeneous population and much of that population binds the resin weakly. Raising the conductivity screens those weak interactions, so more of the population passes into the flow-through and the wash. Raising the protein load lowers the fraction removed, because the antibody competes for binding sites and displaced species can co-elute with it. The two effects are not additive. Their product sets the pool level, so the rise in pool host cell protein with load is larger when the conductivity is low, and platform experience predicts a significant interaction between the two parameters.

The step does not discriminate charge variants or glycosylation variants under platform conditions, and neither class of variant is measured in this study. Both are formed in the production bioreactor and are controlled there (PCR-003). Clearance of residual DNA and of leached Protein A is expected to be independent of the parameters studied. Residual DNA is polyanionic and does not bind the resin under the load conditions used, and leached Protein A is removed largely in the wash, so both clearances follow from the chemistry of the step and not from its operating point.

The material loaded onto the step is the neutralized and filtered pool from the low-pH viral inactivation step. That hold adds aggregate to the intermediate (PCR-006), so the burden this step has to remove is set upstream and is not constant. At nominal process conditions the intermediate carries about 1.9 % HMW, so the study has to characterize the step at an aggregate burden above nominal as well as at nominal, and the worst-case challenge below defines how that is done.

4.2 Quality attributes in scope

This step sets no quality attribute of its own, because every attribute it governs is formed upstream and reduced here. The attributes in scope are therefore clearance attributes, and they are listed in Table 4.1 with their drug-substance acceptance criteria and with the criticality assigned in the risk assessment (RA-001), which follows the continuum of criticality of ICH Q9 (International Council for Harmonisation 2023).

Table 4.1: Quality attributes this step controls or clears, with drug-substance acceptance criteria.

CQA	Category	Acceptance	Criticality	Tool #1
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6
Leached Protein A	process impurity	0–5 ppm	L-M	16

Aggregate determines the design of this study, because it is of high criticality, it is reduced at this step alone, and its acceptance criterion is a drug-substance limit of 5 % HMW by SEC. Host cell protein is of moderate to high criticality and is reduced at three steps. Protein A capture provides the largest single reduction (PCR-005), this step provides a further reduction, and anion exchange provides the last (PCR-008). This step's contribution will be reported both as a clearance factor and as a pool level, and neither figure can be judged in isolation from the other two steps.

Residual DNA and leached Protein A are of low to very low criticality at drug-substance level and are cleared with margin by the train as a whole. They will be measured across the step to confirm the platform expectation stated above, but they are not responses of the designed experiments, because no parameter in scope is expected to change them.

4.3 Risk-based prioritization of parameters

The pre-characterization risk assessment set the scope of this study (RA-001), and each parameter was ranked there for its potential impact on quality attributes and for its potential to interact with other parameters, with the combined score determining the study type. Of the 5 parameters in scope, 4 were assigned to the multivariate design and 1 to univariate assessment.

The ranges to be studied are wider than the routine control ranges, and each characterization range spans between 1.3 and 2.1 times the width of the corresponding normal operating range (Table 6.1). The widening serves two purposes: it places the edges of the studied region far enough outside routine operation that an excursion can be assessed against data, and it leaves room for later movement within the design space without a new study.

The rationale for each range is as follows.

- Protein load will be studied from 10 to 40 g/L resin, against a routine range of 18 to 32, because it is the parameter most likely to move in manufacture, following the harvest titre and the recovery of the two preceding steps.
- Load and wash conductivity will be studied from 3 to 7 mS/cm, a range that covers the buffer preparation tolerance together with the conductivity of the neutralized feed, which varies with the volume of titrant used at the low-pH step.
- Elution buffer pH will be studied from 5.8 to 6.2 pH, a range that is narrow in absolute terms because a step elution is sensitive to small changes in pH, and that is still wider than the control range the commercial skid can hold.
- Elution stop collect will be studied from 0.5 to 1.5 OD desc., which spans collection stops well before and well after the routine point on the descending edge of the peak.
- Elution flow rate will be assessed univariately from 100 to 300 cm/hr, since it acts through residence time and through the pressure drop across the bed and not through the binding chemistry, so no interaction with the other parameters is expected.

5 Materials and methods

5.1 Scale-down model and its qualification

The study will be executed on a scale-down model of the commercial column, qualified against manufacturing-scale data under SOP-1001. The model holds the bed height, the linear velocity, the load in grams per litre of resin and all buffer volumes expressed in column volumes, so that the residence time and the ratio of impurity to available binding capacity match the commercial step. Column efficiency will be verified before each block of runs by plate count and peak asymmetry, and a column outside the acceptance criteria of SOP-2010 will be repacked before use.

Qualification will compare the scale-down model with the commercial-scale process at the set-point condition, and the comparison covers step yield, pool volume, pool aggregate and pool host cell protein, using replicate runs so that means are compared and not single values. The model will be accepted when no statistically significant difference is found in those attributes at alpha = 0.05. The qualification result will be reported in PCR-007 ahead of any characterization result, because every claim in that report depends on it.

Resin used in the study will be within the cycle range qualified under SOP-2008 and the cycle number of each run will be recorded, since resin life-cycle is characterized separately and is not a factor here.

5.2 Operation

The step will be operated as described in SOP-2010, and each run comprises equilibration, load, a post-load wash at the conductivity under study, a step elution, and a strip and sanitization cycle. Product collection starts on the ascending edge of the elution peak at a fixed absorbance threshold and ends at the stop collect set-point on the descending edge. The start threshold is held at the platform value in every run, so that the pool boundary under study is the stop point alone.

Load material will be the neutralized and filtered pool from the low-pH viral inactivation step, prepared as a single pooled lot so that feed variability does not confound the factor effects. The pool will be characterized for aggregate, host cell protein, residual DNA and leached Protein A before execution begins, and those values will be reported in PCR-007 as the study feed composition. Aliquots that cannot be run on the day they are drawn will be held under the storage conditions qualified in SOP-2010.

5.3 Analytical methods

The 4 validated methods that support the study are listed in Table 5.1, and size variants by SEC-HPLC (AMV-3011) is the primary one, reporting the high molecular weight content of both the load and the pool. Host cell protein is measured by ELISA (AMV-3012), residual DNA by qPCR (AMV-3014) and leached Protein A by ELISA (AMV-3016). Accuracy, precision and reportable range are defined in each validation report and are not repeated here.

Table 5.1: Validated analytical methods used in the study.

Reference	Title	Type
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation
AMV-3016	Leached Protein A by ELISA (ppm)	Method validation

Samples will be assayed in the analytical laboratory at the sending site, under the same methods used for release testing. The centre-point replicates carry process and assay variability together, so the pure-error term used in the lack-of-fit test includes the assay contribution. Where an effect is statistically significant but small against that error, PCR-007 will report it as such and will state the practical size of the change over the range studied.

5.4 Sampling plan

Two samples will be taken from every run: the load sample from the prepared pool aliquot, and the product sample from the collected eluate after mixing and before any hold.

Aggregate and host cell protein will be measured on the load and on the pool of every run in both designs, since they are the responses of the fitted models. Residual DNA and leached Protein A will be measured on the centre-point runs and on the runs at the extremes of protein load and conductivity, which is sufficient to confirm the clearance expectation stated above without adding assay burden that supports no model. Step yield will be calculated for every run from the pool volume and the pool concentration.

The two designs carry different numbers of centre points, in that the screening design carries 3 and the response-surface design carries 4, which reflects the greater weight the response-surface analysis places on the pure-error estimate. The whole sampling plan is summarized in Table 5.2.

Table 5.2: Planned sampling and testing per run.

Sample point	Attribute	Method	Runs
Load (pool aliquot)	Aggregate (HMW)	AMV-3011	Every run
Load (pool aliquot)	Host cell protein	AMV-3012	Every run
Load (pool aliquot)	Residual DNA, leached Protein A	AMV-3014, AMV-3016	Centre points and load/conductivity extremes
Product pool	Aggregate (HMW)	AMV-3011	Every run
Product pool	Host cell protein	AMV-3012	Every run
Product pool	Residual DNA, leached Protein A	AMV-3014, AMV-3016	Centre points and load/conductivity extremes
Product pool	Volume and concentration (step yield)	SOP-2010	Every run

5.5 Statistical methods

All models will be fitted on coded factors, using coded levels −1/0/+1, where the centre level is the set-point and ±1 are the edges of the characterization range. The screening design will be analysed by ordinary least squares, with all main effects and all two-factor interactions in the model. An effect is reported as twice the coded coefficient, so that it expresses the change in the response across the full studied range of the factor, and significance will be assessed at alpha = 0.05.

The screening analysis identifies which factors and interactions are active. It does not provide the predictive model. The response-surface design adds the axial runs needed to estimate curvature, and the full quadratic model fitted to that design is the model used for the operating region, for the proven acceptable ranges and for every prediction reported in PCR-007. That distinction between the two models will be stated in the report wherever a model is used.

Model adequacy will be judged on four criteria: a significant model F test; a lack-of-fit F test that is not significant against the pure error from the centre points (Table 5.2); agreement between the adjusted and the predicted coefficient of determination (computed from PRESS); and residual diagnostics showing no structure against the fitted values and no material departure from normality. A model that fails on lack of fit will be augmented before it is used, and PCR-007 will describe the augmentation. A response for which no factor is active will be reported as robust over the ranges studied, and that absence of effect will be retained in the knowledge space as evidence of robustness.

Statements about assurance will be made with the other parameters varying and not held. The proven acceptable range analysis propagates the normal operating ranges of the remaining parameters by Monte-Carlo simulation, using 2,000 draws at each of 81 points across the characterization range of the parameter under analysis.

6 Study design

6.1 Factors, ranges and study type

The design has to do two things: separate the effects of parameters that are known to act on the same attribute, and describe the curvature that a clearance step shows near its limits. The parameters, their set-points, their characterization ranges, their normal operating ranges and their assigned study type are given in Table 6.1. No classification appears there, because classification is an outcome of the study and is assigned in PCR-007.

Table 6.1: Parameters in scope, with the ranges to be studied and the assigned study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Protein load	g/L resin	25	10–40	18–32	multivariate
Load/Wash conductivity	mS/cm	5	3–7	4–6	multivariate
Elution buffer pH	pH	6	5.8–6.2	5.85–6.15	multivariate
Elution stop collect	OD desc.	1	0.5–1.5	0.7–1.3	multivariate
Elution flow rate	cm/hr	200	100–300	150–250	univariate

The 4 multivariate parameters are coded A to D in the designs and in every table of PCR-007, as set out in Table 6.2. Elution flow rate does not appear there, because it is assessed one factor at a time.

Table 6.2: Factor legend for the multivariate designs.

Code	Factor	Unit	Studied in
A	Protein load	g/L resin	screening + RSM
B	Load/Wash conductivity	mS/cm	screening + RSM
C	Elution buffer pH	pH	screening + RSM
D	Elution stop collect	OD desc.	screening + RSM

6.2 Screening design

The screening design is a full two-level factorial in the 4 multivariate parameters with 3 centre points, giving 19 runs (Appendix A). It estimates all 4 main effects and all 6 two-factor interactions without aliasing. A resolution IV fraction would have cost fewer runs, but it would confound the two-factor interactions in pairs, and the interaction between protein load and conductivity is the one result this study cannot leave confounded.

Runs will be executed in random order (SOP-1002), and the centre points will be distributed through the run order so that any drift over the execution window appears in the pure-error estimate and not in a factor effect.

6.3 Response-surface design

The response-surface design is a face-centred central composite in the same 4 parameters (Appendix B). It comprises 16 factorial runs, 8 axial runs at the faces of the cube and 4 centre points, giving 28 runs. The axial runs sit on the faces and not outside the cube, so that no run is executed outside the characterization ranges in Table 6.1. Together the two designs comprise 47 runs, and both will be executed on the same feed pool.

The quadratic terms matter most for aggregate, whose pool level is the load level divided by the clearance the step achieves, so a proportional loss of clearance produces a rise in the pool that is more than proportional. A first-order model would describe that only near the centre of the region, and an operating region has to be defined at its edges.

6.4 Univariate assessment

Elution flow rate will be assessed one factor at a time at the low, centre and high settings of its range, with the 4 multivariate parameters held at their set-points. Each setting will be run in duplicate. Flow is expected to act on the separation only through residence time and pressure drop, and platform experience gives no reason to expect an interaction with the binding chemistry.

The assessment carries a decision rule. If a flow setting changes pool aggregate or pool host cell protein by more than the centre-point variability of the response-surface design, flow will be carried into a follow-up multivariate study, and PCR-007 will state that the region defined here does not cover flow.

6.5 Worst-case aggregate challenge

Executing both designs on one feed pool controls feed variability, and it leaves one question open. It does not show whether the step still meets the aggregate criterion when the material it receives is worse than nominal. Since no downstream step reduces aggregate, that question has to be answered here.

A worst-case challenge will therefore be run separately from the designed experiments. The challenge feed will be prepared by blending the study pool with material of elevated high molecular weight content, generated by extending the low-pH hold of the same intermediate (PCR-006), to an aggregate level at the drug-substance limit of 5 % HMW. That is a conservative bound on any material this step can be asked to process. The intermediate entering the step at nominal conditions carries about 1.9 % HMW (PCR-006), so the challenge loads the step at about 2.6 times the expected burden.

The challenge will be run in triplicate at the corner of the characterized region that the fitted response-surface model identifies as least favourable for aggregate clearance, and once at the set-point condition on the same feed for reference. Its acceptance criterion is given below.

7 Acceptance and decision criteria

The study will be judged against criteria fixed before execution. They fall into three groups: the qualification of the scale-down model, the adequacy of the fitted models, and the quality outcome at the product pool.

The scale-down model criteria are given above. No characterization result will be interpreted until the model has been qualified against manufacturing-scale data.

The model adequacy criteria are the four given above. A response whose response-surface model meets all four will be used for prediction, for the operating region and for the proven acceptable ranges. A response whose model fails one of them will be reported with the failure stated, and any range claim resting on it will be restricted to the conditions actually tested.

The quality criteria are the drug-substance specifications in Table 4.1. Pool aggregate must be at or below 5 % HMW. Pool host cell protein is assessed against the drug-substance limit of 100 ng/mg, and this step is one of three that determine whether that limit is met.

Host cell protein needs a second criterion, because a single step cannot be judged alone against a limit that three steps deliver together. The pool level will be assessed against the drug-substance limit, and it will also be assessed as a clearance factor relative to the measured load level. Where the pool level exceeds the drug-substance limit somewhere in the studied region, that condition is not by itself a failure of the step. It becomes a constraint on the operating conditions of the anion exchange step, and the two steps will be reconciled in PCMR-001. One alternative was considered and rejected. Setting an arbitrary in-process limit on host cell protein at this step would be simpler to administer, but it would constrain the operating region without evidence that the constraint is needed.

The worst-case challenge has a single criterion. Pool aggregate must be at or below 5 % HMW in every challenge run, with the feed at the drug-substance limit.

Step yield carries no drug-substance specification and is characterized as a process performance attribute. A yield loss at any studied condition will be reported together with the conditions that produce it.

The study will be reported as complete when the scale-down model has been qualified, both designs have been executed and analysed, the univariate assessment and the worst-case challenge have been executed, and every parameter in Table 6.1 has been classified under SOP-4001. If a design has to be repeated, the reason, the disposition of the affected data and the re-executed design will be described in the deviations section of PCR-007.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range will be determined for each parameter and each governed attribute. The design space is multivariate and a proven acceptable range is not, so the two will be reported side by side and neither replaces the other.

The acceptance basis is the drug-substance specification for the attribute (Table 4.1). This step has no viral clearance response, so no criterion is back-calculated from a cumulative requirement, and every criterion used in this analysis is a direct specification limit.

Two analyses will be run for each parameter and each attribute. The first holds the remaining parameters at their set-points and scans the parameter across its characterization range, and it gives the range over which the fitted model predicts an in-limit result under otherwise nominal operation. The second varies the remaining parameters within their normal operating ranges by Monte-Carlo simulation of the fitted model, and it gives the range that survives routine co-variation of everything else.

The decision rule is the same for both analyses. The proven acceptable range is the contiguous interval around the set-point over which the predicted response meets the acceptance criterion. Where the two analyses agree across the whole characterization range, the studied range is proven acceptable and robust to co-variation, and the binding constraint on operation is the multivariate design space and not the range of any single parameter. Where the second analysis is narrower, PCR-007 will name the attribute that binds it and will give the remaining margin to the set-point.

Each governed attribute will be presented with a plot of the response against its governing parameter, with the acceptance limit drawn, the acceptable region shaded, and the set-point and normal operating range marked. The two analyses will appear as the two panels of that plot.

Both analyses rest on the fitted response-surface model, so both are bounded by the characterization ranges in Table 6.1 and by the validity of the scale-down model. Neither extrapolates beyond the studied region. Neither replaces the confirmation of performance at commercial scale, which is covered by PTP-001.

9 Data management and integrity

Study records will be maintained under the site data-integrity procedures and will meet the ALCOA+ expectations. Run execution will be recorded in the electronic laboratory notebook, analytical results in the laboratory information management system, and the statistical analysis in a version-controlled script archived with the study record (SOP-1002). Raw instrument files will be retained without modification, and every derived value in PCR-007 will be traceable to the file it came from. A second qualified person will review the analysis before the report is issued.

10 Roles and responsibilities

Responsibilities are allocated as shown in Table 10.1. Process Development at the sending site owns the study and the report. Analytical Development owns the methods and the assay data, Statistics owns the design and the analysis, and Quality Assurance approves this plan and the report that follows it.

Table 10.1: Roles and responsibilities for the study.

Function	Responsibility
Process Development / MSAT (sending site)	Study ownership; scale-down model; execution; authorship of PCR-007
Analytical Development	Validated methods; sample testing; assay data review
Statistics	Design generation and randomization; model fitting; region and range analysis
Manufacturing Science (receiving site)	Review of commercial-scale equivalence and of the proposed ranges
Quality Assurance	Approval of this plan and of PCR-007; deviation oversight

11 Deliverables and schedule

The deliverable is PCR-007. It will report the executed designs, the fitted models, the operating region, the proven acceptable ranges, the parameter classifications and any departure from this plan.

The study is executed in the sequential phases listed in Table 11.1. No calendar dates are fixed here, since execution is scheduled against material availability. The univariate assessment may run in parallel with the response-surface design, because it uses the same feed pool and does not depend on the multivariate result.

Table 11.1: Study phases and their outputs.

Phase	Output
Scale-down model qualification against manufacturing-scale data	Qualification data package
Feed pool preparation and characterization	Study feed composition
Screening design execution and analysis	Active factors and interactions
Response-surface design execution and analysis	Predictive model for each response
Univariate assessment of elution flow rate	Flow decision under the rule stated above
Worst-case aggregate challenge	Challenge result against the aggregate criterion
Region, range and classification analysis, and reporting	PCR-007

12 Risks and assumptions

Three assumptions carry the study, and each is stated here so that PCR-007 can be read against them.

The first assumption is that the scale-down model represents the commercial column. The qualification described above tests that assumption at the set-point condition only. Operating regions defined at scale-down are not confirmed at scale at the edges of their ranges, and the strategy for confirming performance at the receiving site is defined in PTP-001.

The second assumption is that a single pooled feed represents the material the step will receive. Aggregate and host cell protein in the feed vary with upstream operation. The study addresses that variation in two ways, in that the load level of both attributes is measured for every run, and the worst-case challenge tests the step at an aggregate burden above nominal. Variation in feed host cell protein is not challenged here, and the range the capture step delivers is reported in PCR-005.

The third assumption is that charge variants and glycosylation variants are unaffected by the step. This rests on the platform experience and the mechanism described above, and it is not tested in this study.

Two risks could delay the study or weaken its conclusions. Insufficient inactivated pool would force execution onto more than one feed lot, which would add a blocking factor to both designs and reduce the precision of the estimated effects. Assay variability at the low end of the host cell protein range could widen the confidence intervals enough that a small effect cannot be resolved, in which case PCR-007 will report the size of the effect the study was able to detect.

13 Approvals

This plan takes effect when the approval record in Table 1 is complete, and execution may not begin before then. Any change to the design, to the ranges or to the acceptance criteria after approval requires a revision of this plan. Any departure during execution will be recorded at the time and assessed in PCR-007.

14 Appendices A–B — Planned design matrices

The two planned design matrices are given below in coded form, and the natural-unit setting for each coded level follows from the characterization ranges in Table 6.1. No response data appear in either matrix, because none exist at the time of approval.

14.1 Appendix A — Planned screening design matrix

Table 14.1: Planned screening design matrix, coded (factors A to D per Table 6.2).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	center	0	0	0	0
18	center	0	0	0	0
19	center	0	0	0	0

14.2 Appendix B — Planned response-surface design matrix

Table 14.2: Planned response-surface design matrix, coded (factors A to D per Table 6.2).

Run	Type	A	B	C	D
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

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